A Review of Wavelet Transform based Techniques for Denoising Latent Fingerprint Images

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***Abstract*—Authentication systems robustness can be affected by the fingerprint image quality. Fingerprint image denoising is essential for better performance of any authentication system. In this paper, most recent wavelet transform based techniques for fingerprint image denoising are reviewed. It is observed that there are four important components of wavelet transform based denoising techniques. These important components include wavelet filter, thresholding rule, threshold value computation method and the level of decomposition. The stationary wavelet transform and the weighted median are recommended over conventional discrete wavelet transform and median estimator.**

Keywords— Denoising, Fingerprints, Wavelets, threshold, Biometrics.

I. INTRODUCTION

Biometrics refers to the measurement and statistical analysis of people's physical and behavioral characteristics [1]. Biometric technologies are widely used in various applications for authentication purposes. Biometrics traits include voice, gait, face and fingerprints. Among these traits, fingerprints are the most used because of their uniqueness, performance and universality [2].

There are several ways of acquiring fingerprints; for examplar fingerprints these include cameras, contact fingerprint sensors and contactless fingerprint sensors. For latent fingerprints they include powders, chemicals, dyes, tape, alternative light sources and contactless sensors The traditional contact-based techniques for lifting latent fingerprints have a few limitations. Using chemicals or dyes alters the crime scene, destroying any possibilities of secondary analysis of the sample and using tape to lift the fingerprint destroys the sample. In efforts to preserve crime scenes, forensics is moving towards acquiring latent fingerprints using contactless techniques which use advanced sensors such as optical coherence tomography (OCT) [3], chromatic white light sensors [4], alternative light sources [5], infrared sensors [6] and ultraviolet sensors [7]. However, some of these sensors suffer from producing noisy images.

Noise is any unwanted modification of an original image or signal. It may originate from scanner hardware, sensor or camera or the surrounding environment [2]. Noise degrades the quality of an image by interfering with the original signal, which results in the variation of pixels values [8]. Noise signals may be additive or multiplicative in data.

Before any processing is done on an image, denoising is necessary. Valuable information hidden by a noise signal is unmasked during denoising. Conventional linear filters are used in denoising images [2]. However, these filters are not effective at preserving important features as the image is blurred [8]. Specific noise types require specific types of filters. For example, Impulse noise can be effectively denoised using a median filter or morphological filter [9, 10] , while Gaussian noise can be effectively denoised using median filter [9, 11], and Speckle noise can benefit from optimized Bayesian non-local mean (OBNLM) filtering [12], Wiener filtering and wavelet transform techniques [13, 14]. Latent fingerprints captured using OCT suffer from a lot of speckle noise which needs to be denoised to render the fingerprints usable by automatic recognition algorithms.

Techniques that operate in a frequency domain are powerful tools for image de-noising. The Fast Fourier transform (FFT) is used to transform an image into the frequency domain. However, the Fourier transform does not represent abrupt changes efficiently [15]. This is because it represents data in the form of sine waves which are not localized in time and space. A new class of functions called wavelets which are well localized in time and frequency can be used to analyse signals and images with sharp changes accurately. Fig. 1 shows the typical fingerprint authentication system, while Fig. 2 shows the denoising process that is highlighted by two lines in Fig. 1. This paper reviews wavelets based techniques that have been appied on fingerprint images in the past 8 years.

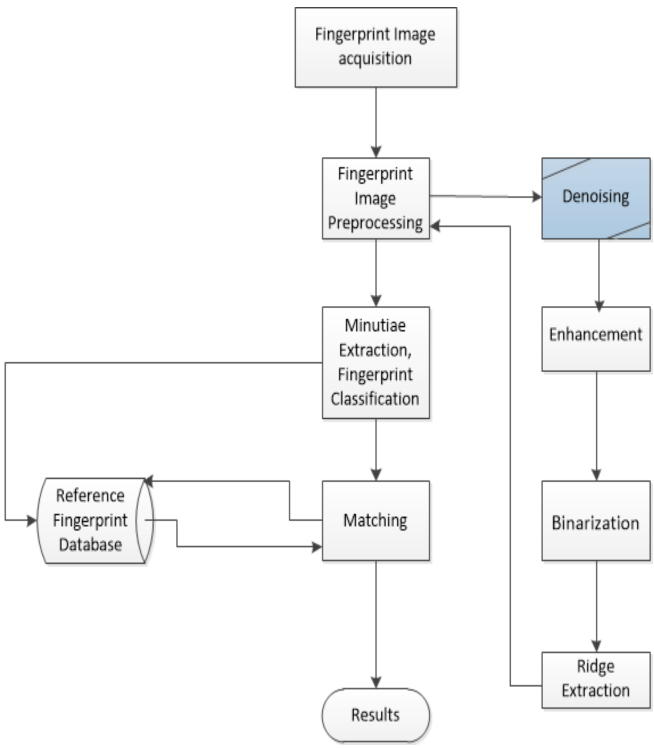

Fig. 1: Typical fingerprint authentication systems adopted from [16].

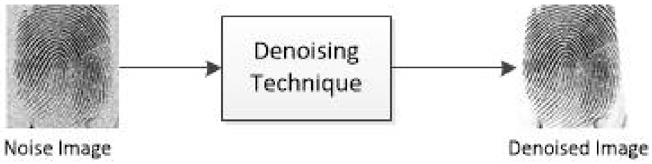

Fig. 2: Typical fingerprint image denoising process. Image is taken from the NIST database [17].

II. Wavelets Concept

A. Wavelets decomposition

If an image is convolved with low pass and high pass filter in the vertical and horizontal direction, four sub-images of the original image are formed [18]. The sub-image encompasses the full image, but at different resolutions and containing different components of the original image. This includes a sub-image of low frequency in both the horizontal and vertical direction (LL), sub-image with low horizontal frequency but high vertical frequency (LH), sub-image with high horizontal frequency but low vertical frequency (HL) and (HH) which has high frequency in both horizontal and vertical directions. Fig. 3 shows the decomposition skeleton of the image up to level 2, where LL is the approximation image (i.e. the enhanced output), HL, LH and HH represent image details in the horizontal, vertical and diagonal direction respectively, the subscript represents the decomposition level. In fig. 4 a fingerprint image was decomposed up to level 2 using Haar wavelet filter.

<table>
<tr><td>LL_2</td><td>HL_2</td><td rowspan="2">HL_1</td></tr>
<tr><td>LH_2</td><td>HH_2</td></tr>
<tr><td colspan="2">LH_1</td><td>HH_1</td></tr>
</table>

Fig. 3: Image wavelet decomposition.

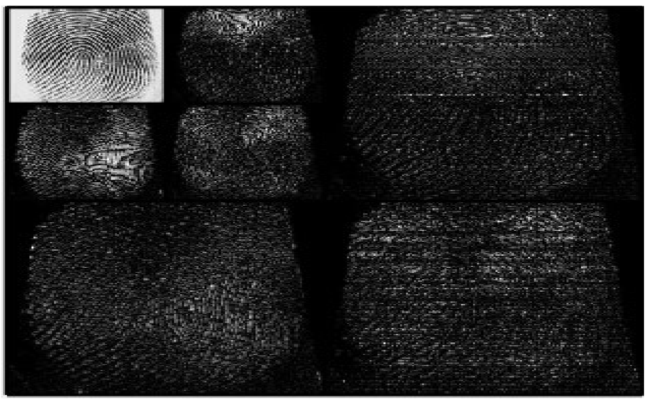

Fig. 4: Level 2 wavelet decomposition. Fingerprint image taken from the NIST database [17].

B. Soft and hard Thresholding

Image thresholding is the method of partitioning an image into foreground and background [19]. It is simple and mostly used where ever two regions need to be classified or distinguished. In wavelet transform, thresholding is done on one wavelet coefficients at a time [19]. Each coefficient is compared to the threshold value depending on the thresholding rules, which are hard and soft thresholding. All coefficients in hard thresholding are either zero or greater or equal to the threshold value, as shown in (1) [20, 21].

$$f(x)_h = \begin{cases} x, & if\ |x| \geq T \\ 0, & if\ |x| < T \end{cases} \tag{1}$$

where, *T*, is the threshold value, for soft thresholding, wavelet coefficients are reduced to a threshold value as shown in (2) [20, 21].

$$f(x)_s = \begin{cases} x - T, & if\ |x| > T \\ 0, & if\ |x| \leq T \\ x + T, & if |x| < -T \end{cases} \tag{2}$$

Leavline *et al.* [15] conducted a study of various wavelet thresholding techniques for denoising. The study includes VisuShrink, BayesShrink, normalShrink, modified BayesShrink, Bivariate shrink, SureShrink, waveShrink. These techniques are often used to find the optimal threshold value T, as it controls the performance of the denoising algorithm. Thresholding techniques such as VisuShrink, BayesShrink, NeighShrink and normalShrink have been used by various researchers [14, 20, 22] for image denoising.

III. Detailed Leterature Review

The following subsections describe different wavelet techniques used by various researchers for noise removal on fingerprint images.

A. Yinping Adaptive Threshold

The work done by Yinping *et al*. [18] proposes an adaptive threshold method based on the wavelet transform for fingerprint denoising.

1) Database: The work uses images from the NIST special database [17].

2) Algorithm: The proposed method uses the same principles of wavelets decomposition. The difference is how the threshold value T is computed. The threshold value T is critical when using wavelets. Conventionally VisuShrink and BayesShink are used to compute the optimal threshold. The VisuShrink threshold (or Universal threshold) is given by:

$$T = \sigma\sqrt{2logN} \tag{3}$$

Where σ, is the noise standard deviation and N is the number of high frequency coefficients found by applying a discrete wavelet transform (DWT) on an image. Bayes Shink threshold is defined as follows:

$$T = \frac{\sigma_{noise}^2}{\sigma_{signal}} \tag{4}$$

Where σ_{noise}^2, is the estimate of noise signal variance and σ_{signal} is the estimate of image signal standard deviation.

Based on VisuShrink and BayesShrink threshold, Yinping *et al.* [18] defined an adaptive threshold Shrink, which computes an adaptive threshold of different sub-images (LL) using the different directions of coefficients details (LH, HL, HH). These are achieved through the following steps:

Let the original image be represented by $F(x,y)$, where $x,y = 1,2,\dots,M$. The noisy image is defined as:

$$G(x,y) = F(x,y) + \Phi(x,y) \tag{5}$$

where $\Phi(x,y)$ is the noise function. Applying the wavelet transform, the noisy image function becomes:

$$W_G = W_F + W_\Phi \tag{6}$$

Where W_G, W_F and W_N are the wavelet coefficients of the image containing noise, the original image and noise signal respectively. The standard deviation of the noise signal coefficients is estimated using the median estimator with diagonal detail coefficients (HH) as follows:

$$\sigma_{W_\Phi} = \frac{median(|HH_i|)}{0.6745} \tag{7}$$

Where $i = 1,2,\dots,k$ k represents wavelet decomposition layers. The signal variance of the image containing noise is computed using the coefficients of each direction detail. These coefficients are as follow:

Let $D = 1,2,3$ represent horizontal, vertical and diagonal details respectively, then $W_G(1,i) \in LH_i$, $W_G(2,i) \in HL_i$ and $W_G(3,i) \in HH_i$. These direction details are used to estimate the noisy $(G(x,y))$ image signal noise variance using (8):

$$\sigma_{W_G}^2(D,i) = \frac{1}{N(i)^2}\sum_{x=1}^{N(i)}\sum_{y=1}^{N(i)} W_G(D,i) \tag{8}$$

analogously the noisy image in terms of variance is defined as,

$$\sigma_{W_G}^2 = \sigma_{W_F}^2 + \sigma_{W_\Phi}^2 \tag{9}$$

where $\sigma_{W_F}^2$ and $\sigma_{W_\Phi}^2$ represent the variance of the original and noise signal respectively. The original image variance is then defined as:

$$\sigma_{W_F}(D,i) = \sqrt{max(\sigma_{W_G}^2(D,i) - \sigma_{W_\Phi}^2(i),0)} \tag{10}$$

Where $\sigma_{W_\Phi}^2(i)$ is noise at different layers. Finally, the adaptive threshold Shrink is given by:

$$T(D,i) = \begin{cases} \dfrac{\sigma_{W_\Phi}^2(i)}{\sigma_{W_F}(D,i)}, & if \quad \sigma_{W_F}(D,i) \neq 0 \\ max\left(W_{x,y}(D,i)\right), & if \ \sigma_{W_F}(D,i) = 0 \end{cases} \tag{11}$$

The noisy fingerprint image is then denoised using the procedure in algorithm 1.

Algorithm 1: Denoising procedure of adaptive threshold Shrink.

Input : Noisy fingerprint image; **Output**: Denoised Image

1. Add noise with $\sigma = 10$, 15 and 20.
2. Decompose the noisy image using forward DWT.
   a. Estimate the noise signal standard deviation (σ_{W_Φ}), HH_i coefficients $(i = 1,2,\dots,k)$.
   b. Calculate the varience $(\sigma_{W_G}^2)$ of the noisy image.
   c. Calculate the standard deviation (σ_{W_F}) of the original image.
3. Threshold the wavelet coefficients in detail subband using the adaptive threshold Shrink.
4. Reconstruct using IDWT to get the denoised image.
5. Evaluate the performance using image metrics quality measure (PSNR).

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3) Experimental Results: Noise with a standard deviation of $\sigma = 10$, 15, and 20 was added on grayscale images of size 256 by 256. The proposed adaptive threshold outperformed the conventional wavelet thresholding techniques by improving the Peak Signal to Noise Ratio (PSNR) by 3.3 % for $\sigma = 10$, 4.7% for $\sigma = 15$ and 5.6% for $\sigma = 20$.

B. Noise Adaptive wavelet thresholding (NAWT)

The work done by Zaki *et al.* [13] proposes a noise adaptive threshold wavelet transform (NAWT) algorithm for speckle noise removal from fingerprint images which utilises the noise contrast at different wavelet subbands.

1) Database: Self acquired fingertip images using spectral domain optical coherence tomography (SD-OCT). The SD-OCT system has a superluminescent diode (SLD1325) as the broadband light source.

2) Algorithm: Their proposed method includes image acquisition, wavelet transform, threshold value computation for all subbands $(LH_i, HL_i$ and $HH_i)$, soft thresholding and inverse wavelet transform. The threshold value for all subbands is computed using (12):

$$T(\sigma_X) = \frac{\sigma_W^2}{\sigma_X} \tag{12}$$

Where σ_W^2 represents the noise variance for the subband W (W is all the different details: Horizontal, vertical and diagonal at different levels) and σ_X is the signal variance of the noisy image. In this technique, a structure less sample with homogeneous optical properties was used to estimate σ_W^2 and this is done because the speckle noise in OCT images depends on the imaging systems rather than the sample.

3) Experimental results: The method was tested with three different samples of which two of them are non–fingerprints images. The results for the fingertip sample of the technique that was proposed outperformed the Gaussian filter (GF) and traditional wavelet transform (WF) by having

SNR of 20.59 while GF and WF have 18.99 and 17.99 respectively. The traditional WT outperformed NAWT in terms of feature preserving β. The value of β = 0.94 was obtained for WF which outclassed NAWT and GF by 1% and 4% respectively.

C. Stationary wavelet transform

Iqbal [14] studied stationary wavelet transform to denoise fingerprint images using four different wavelet filters including Haar, Daubechies (db4), Coiflets (coif2) and Biorthogonal (bior3). For each wavelet, the optimal threshold value (T) was computed using BayesShrink, Visu Shink, normalShrink and NeighShrink [15].

1) Database: The fingerprint images were taken from the NIST database [17].

2) Algorithm: Iqbal used VisuShrink, BayesShrink, normalShrink and NeighShrink to compute the optimal threshold for each sub-image. The major difference in Iqbal's [14] work is that stationary wavelet transform is used instead of a DWT.

3) Experimental results: Their results showed that BayesShrink at level 2 combined with Haar wavelet filter remove Gaussian and speckle noise successfully. NeighShrink at level 1 combined with Haar wavelet filter successfully removed salt and pepper noise.

D. Modified universal threshold

The work done by Sasirekha *et al.* [23] proposes a Gaussian noise removal technique that uses stationary wavelet transform and modification of the universal threshold.

1) Database: Fingerprint database FVC2002 [24] was used to test the technique.

2) Algorithm: Their proposed method modifies the VisuShrink threshold method also known a universal threshold (3) and median estimator. They replaced the value 2 by the golden ratio value (1.618), where (3) becomes:

$$T = \sigma\sqrt{1.618logN} \qquad (13)$$

The weighted sum was used to calculate the coefficients of the diagonal subband (HH), given by:

$$W(x,y) = \frac{1}{\exp(HH(x,y))} \qquad (14)$$

then $HH1$ is given by,

$$HH1(x,y) = W(x,y) * HH(x,y) \qquad (15)$$

Where x and y are the coordinates in the HH subband. The weighted diagonal HH1 subband is the product of weight function and HH subband (15). Hence the noise variance is given by:

$$\sigma = \frac{median|HH1|}{0.6745} \qquad (16)$$

The noisy fingerprint image is then denoised using the procedure in algorithm 2 adapted from [23]:

Algorithm 2: Denoising algorithm of the modified universal thresholdusing golden ratio and weighted median.

Input: Noisy image; **Output**: Denoised Image.

1. Add Gaussian Noise to the image.
2. Decompose the noisy image using forward startionary wavelet transform (SWT).
   a. Compute weights for the coefficients in the HH subband using the weight function.
   b. Find the product of weight function with HH to get diagonal subband HH1.
   c. Calculate noise variance using diagonal subband HH1.
3. Threshold the wavelet coefficients in detail subband using the modified universal threshold.
4. Reconstruct using inverse SWT to get the denoised image.
5. Evaluate the performance using image metrics quality measure (MSE, RMSE, PSNR and SNR).

3) Experimental results: The modified universal threshold outperformed the median filter, M3 filter and traditional universal threshold. Tables I-VI shows the results. Table I and II show that M3 outperforms median filter in both noise levels (0.001 and 0.003). Table III and IV show that Symlets (sym4) and db2 were optimal wavelet filters for soft and hard thresholding rule respectively for traditional universal threshold. The results for the modified universal threshold are shown in Table V and VI for the noise level of 0.001 and 0.003 respectively. The wavelets filters sym4 and DB4 are the optimal filters for soft and hard thresholding value respectively.

TABLE I. MEDIAN AND M3 FILTER (NOISE LEVEL = 0.001)

	MSE	RMSE	PSNR	SNR
Median	229.9637	15.06569	24.66274	2.784469
M3	140.7967	11.79751	26.7798	2.73094

TABLE II. MEDIAN AND M3 FILTER (NOISE LEVEL = 0.003)

	MSE	RMSE	PSNR	SNR
Median	299.1854	17.22912	23.47392	2.917146
M3	202.4989	14.18667	2515389	2.808041

TABLE III. OPTIMAL TREDITIONAL THRESHOLD WITH SOFT AND HARD THRESHOLD (NOISE 0.001).

Rule	Filter	MSE	RMSE	PSNR	SNR
SOFT	SYM4	74.70805	8.565915	29.58568	1.097452
HARD	DB2	41.12677	6.359209	32.16953	0.394085

TABLE IV. OPTIMAL TREDITIONAL THRESHOLD WITH SOFT AND HARD THRESHOLD (NOISE 0.003).

Rule	Filter	MSE	RMSE	PSNR	SNR
SOFT	SYM4	146.6961	12.04885	26.59025	1.417269
HARD	DB2	100.833	9.98619	28.22458	0.806413

TABLE V. OPTIMAL MODIFIED UNIVERSAL THRESHOLD WITH SOFT AND HARD THRESHOLD (NOISE 0.001).

Rule	Filter	MSE	RMSE	PSNR	SNR
SOFT	SYM4	1.747893	1.321577	45.74625	0.13033
HARD	DB2	0.150877	0.38788	56.40315	0.00137

TABLE VI. OPTIMAL MODIFIED UNIVERSAL THRESHOLD WITH SOFT AND HARD THRESHOLD (NOISE 0.003).

Rule	Filter	MSE	RMSE	PSNR	SNR
SOFT	SYM4	1.38164	45.57296	0.122724	0.13033
HARD	DB2	00.088247	0.296831	58.72143	0.001019

IV. DISCUSSION

From the literature [18, 13, 23, 14, 19] it is observed wavelet based denoising techniques for fingerprint images have four important components. The first component is the wavelet filter (haar, db2, sym4, etc.)– for example haar is the simplest of all the wavelet filters, memory efficient and computational cheap, on the other hand db2 is computational expensive since it uses overlapping windows thus the results reflect all changes between pixel intensities– this means db2 is smoother than haar which suggest that it may be good for images corrupted with speckle noise. The second component is thresholding rule which is either soft or hard thresholding. The third component is the level of decomposition– higher levels may distort the image last component is the threshold value computation– a very small threshold may fail to remove some noise coefficient while a larger one may blur some important information. Table VII summaries the techniques discussed in the previous section in terms of these four important components.

TABLE VII. SUMMARY OFT THE TECHNIQUES DISCUSSED IN SECTION III IN TERMS OF FOUR CRUCAIL CHOICES..

Author	Noise removed	Wavelet filter	Decom-position Level	Threshold Rule
Yinping *et al.* [18]	σ = 10, 15, 20	db4	Level 3	soft
Zaki *et al.* [13]	Speckle noise	sym4	Level 4	soft
Iqbal [14]	Speckle, Gaussian and Salt & pepper noise	haar	Level 2	soft
Sasirekha *et al.* [23]	Gaussian Noise	Sym4 and ***DB2***	Level 2	Soft and ***Hard***

The denoising techniques reviewed in this work may be further improved by using a stationary wavelet transform (SWT) instead of conventional discrete wavelet transform (DWT). The DWT has the inefficiency of not being translational invariant and SWT overcome that inefficiency. The noise in the HH subband should be estimated by weighted function. The weighted median estimator is computed using the classical weighted function (W) [25, 23] given by (14). The noise variance is then computed from $HH1$ which is the weighted diagonal subband (horizontal high-frequency details) as given in (16). The weighed median may give a better estimation of noise variance compares to traditional median estimator, which is used to compute the noise signal.

V. CONCLUTION AND FUTURE WORK

The wavelets based techniques for fingerprint denoising have been reviewed in this paper. It is observed that when denoising images using wavelet transform there are four important components that needs to be right. These components include threshold rule, wavelet filter, level of decomposition and threshold computation. It is also seen that wavelet-based denoising techniques are distinguishable by their way of calculating threshold value. Future work will propose the adaptive denoising method that uses SWT and weighted median and compare it with the denoising techniques reviewed in this paper. The OCT latent fingerprint images will be used as dataset.

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REFERENCES

[1] R. Nimkar and A. Mishra, "Fingerprint segmentation algorithms: A literature review," *International Journal of Computer Applications*, vol. 95, no. 5, 2014.

[2] A. Katiyar and V. Santhi, "Region based speckle noise reduction approach using fuzzy techniques," in *2017 International Conference on Intelligent Computing, Instrumentation and Control Technologies (ICICICT)*, pp. 673–676, IEEE, 2017.

[3] S. K. Dubey, D. S. Mehta, A. Anand, and C. Shakher, "Simultaneous topography and tomography of latent fingerprints using full-field swept-source optical coherence tomography," *Journal of Optics A: Pure and Applied Optics*, vol. 10, no. 1, p. 015307, 2008.

[4] M. Leich, S. Kiltz, J. Dittmann, and C. Vielhauer, "Non-destructive forensic latent fingerprint acquisition with chromatic white light sensors," in *Media Watermarking, Security, and Forensics III*, vol. 7880, p. 78800S, International Society for Optics and Photonics, 2011.

[5] P. K. A. Dakshinamurthy S, "Study on efficiency of alternate light source for detection of latent fingerprints," 2018.

[6] N. J. Crane, E. G. Bartick, R. S. Perlman, and S. Huffman, "Infrared spectroscopic imaging for noninvasive detection of latent fingerprints," *Journal of forensic sciences*, vol. 52, no. 1, pp. 48–53, 2007.

[7] N. Saitoh and N. Akiba, "Ultraviolet fluorescence imaging of fingerprints," *The Scientific World Journal*, vol. 6, pp. 691–699, 2006.

[8] S. Kaur, "Noise types and various removal techniques," *International Journal of Advanced Research in Electronics and Communication Engineering (IJARECE)*, vol. 4, no. 2, pp. 226–230, 2015.

[9] M. A. Koli, "Review of impulse noise reduction techniques," *International Journal on Computer Science and Engineering*, vol. 4, no. 2, p. 184, 2012.

[10] C. Anjanappa and S. HS, "Development of mathematical morphology filter for medical image impulse noise removal," in *2015 International Conference on Emerging Research in Electronics, Computer Science and Technology (ICERECT)*, pp. 311–318, IEEE, 2015.

[11] K. Kanagalakshmi and E. Chandra, "Performance evaluation of filters in noise removal of fingerprint image," *2011 3rd International Conference on Electronics Computer Technology*, vol. 1, pp. 117–121, 2011.

[12] L. N. Darlow, S. S. Akhoury, and J. Connan, "A review of state-of-the-art speckle reduction techniques for optical coherence tomography fingertip scans," in *Seventh International Conference on Machine*

Vision (ICMV 2014), vol. 9445, p. 944523, International Society for Optics and Photonics, 2015.

[13] F. Zaki, Y. Wang, H. Su, X. Yuan, and X. Liu, "Noise adaptive wavelet thresholding for speckle noise removal in optical coherence tomography," *Biomed. Opt. Express*, vol. 8, pp. 2720–2731, May 2017.

[14] N. Iqbal, "Reduction of Noise from Fingerprint Images using Stationary Wavelet Trasnform," *International Journal of Engineering Works (ISSN: 2409-277)*, vol. 4, pp. 104–108, Dec. 2017.

[15] E. J. Leavline, S. Sutha, and D. A. A. G. Singh, "Wavelet domain shrinkage methods for noise removal in images: A compendium," *International Journal of Computer Applications*, vol. 33, no. 10, pp. 28–32, 2011.

[16] X. Liang and T. Asano, "A linear time algorithm for binary fingerprint image denoising using distance transform," *IEICE TRANSACTIONS on Information and Systems*, vol. 89, no. 4, pp. 1534–1542, 2006.

[17] http://fingerprint.nist.gov

[18] M. Yinping and H. Yongxing, "Adaptive threshold based on wavelet transform fingerprint image denoising," in *Proceedings of the 2012 International Conference on Computer Science and Electronics Engineering - Volume 03*, ICCSEE '12, pp. 494–497, IEEE Computer Society, 2012.

[19] P. Hedaoo and S. S. Godbole, "Wavelet thresholding approach for image denoising," *International Journal of Network Security & Its Applications (IJNSA)*, vol. 3, no. 4, pp. 16–21, 2011.

[20] A. Al Jumah, "Denoising of an image using discrete stationary wavelet transform and various thresholding techniques," *Journal of Signal and Information Processing*, vol. 4, no. 01, p. 33, 2013.

[21] H. Nasif, "Wavelet denoising," 04 2016.

[22] P. Gupta and A. Garg, "Image denoising using bayes shrink method based on wavelet transform,"

[23] K. Sasirekha and K. Thangavel, "A novel wavelet based thresholding for denoising fingerprint image," in *2014 International Conference on Electronics, Communication and Computational Engineering (ICECCE)*, pp. 119–124, IEEE, 2014.

[24] http://bias.csr.unibo.it/fvc2000/databases.asp

[25] C.-y. Wang, L.-l. Li, F.-p. Yang, and H. Gong, "A new kind of adaptive weighted median filter algorithm," in *2010 International Conference on Computer Application and System Modeling (ICCASM 2010)*, vol. 11, pp. V11–667, IEEE, 2010.